

# Individual-based modelling of black bear (*Ursus americanus*) foraging in Whistler, BC: Reducing human-bear interactions

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## ABSTRACT

Human-black bear interactions have a long standing history involving extermination of bears, damage to human property, and, in some cases, injury or loss of human life. Much work has been done in the field of bear management to reduce the number of human-black bear interactions, including aversive conditioning, relocation, resource management, etc. However, determining which strategies are most effective is challenging and costly in terms of both time and fiscal resources. We approached this problem using an individual-based modelling technique that allows for the examination of multiple different bear management strategies at minimal cost. We tested several different types of bear management strategies (BMS) and bear management spatial configurations (BMC) using the community of Whistler, BC as a case study. Our results indicate that the optimal BMC depends on the BMS used, however, all implementations of bear management resulted in a decrease in the number conflict bears. Models of this type could be used to guide future conservation efforts in similar communities seeking to reduce conflicts between humans and bears.

## 1. Introduction

It has long been acknowledged that human-wildlife interactions are expected to increase with urban growth and development (Woodroffe et al., 2005; Hristienko and McDonald Jr, 2007). As we continue to expand into wildlife habitat it is reasonable to expect the number of conflicts with wildlife to increase, thus creating a demand for mitigation strategies (Hostetler et al., 2009). A prominent example of such conflicts is that between humans and American black bears (*Ursus americanus*). Black bears have experienced dramatic losses in habitat range (McLean and Pelton, 1990; Schoem, 1990) as well as increased involvement in human-bear conflicts (Beckmann and Berger, 2003; Gore et al., 2005; Zack et al., 2003).

Black bears are abundant throughout North America with an estimated total population of 750,235 to 917,650 individuals (Hristienko and McDonald Jr, 2007). A conflict bear is here defined as a bear that

acts on its learned behaviour to such an extent that it produces a threat to human safety and property when seeking human food and/or garbage. Conflict bears are a result of specific types of human-bear interactions, in particular, overexposure of bears to anthropogenic food sources, garbage, and other attractants. Conflict bears typically seek out anthropogenic food and resources, and sometimes damage property in order to gain access to these resources. Any bear can become food conditioned and habituated towards humans to varying degrees, both of which lead to higher likelihood of more conflicts with that bear (Greenleaf et al., 2009; Mattson et al., 1992; Peine, 2001). Consequently, both the prevention and handling of conflict bears are a major focus of modern wildlife management (Gniadek and Kendall, 1998; Gunther, 1994; Herrero et al., 2005).

Historically, lethal control has been implemented as a control measure for human-black bear conflict, but this approach has proved unsatisfactory as human-black bear conflicts continue to occur. As

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human populations increase, it is expected that bear populations will increase as well due to increased availability of high quality anthropogenic food and garbage (Gore et al., 2006; Hristienko and McDonald Jr, 2007; Worthy and Foggin, 2008; Baruch-Mordo et al., 2011; Charoo et al., 2011). It has already been noted that bear populations are increasing in numbers (Garshelis and Hristienko, 2006; Spencer et al., 2007). In British Columbia the estimated population of approximately 140,000 bears results in approximately 10,000 complaints per year, with a trend of complaints increasing (Spencer et al., 2007). Even though bears are known to associate humans with risk (Beckmann and Berger, 2003), conflicts continue to occur.

The continued conflicts have led to the development of several non-lethal management strategies, such as removal of attractants, the installation of animal-proof waste bins, and the use of aversive conditioning and relocation (Hopkins III and Kalinowski, 2013). While non-lethal bear management strategies show promise, they also have shortcomings. For example, while translocation has been used for many bear species, it is relatively expensive and often unsuccessful (Linnell et al., 1997; Riley et al., 1994) with many translocated bears finding their way back to their original home ranges. In one study, only 15% of the wildlife agencies consulted reported it as an effective tool even though over two thirds of them used translocation (Spencer et al., 2007). Aversive conditioning is frequently only a short-term solution requiring extensive resources (Linnell et al., 1997; Baruch-Mordo et al., 2009).

There is evidence that limiting access to anthropogenic food through attractant removal is key to reducing human-bear conflict (Spencer et al., 2007), though it is highly unlikely that all attractants in an urban area can be removed. Attractants include, for example, fruits and vegetables grown in backyard gardens, coolers left unattended, and waste foods found in garbage. In a previous study of conflict types, garbage/food related conflicts were the most common (Spencer et al., 2007), while another survey study in Florida had 50.3% of participants reporting bears accessing garbage (Pienaar et al., 2015). One possible mechanism for implementing attractant removal is to enforce the use of bear-proof garbage bins in both public and residential areas. In a plethora of national parks, wildlife-resistant waste containers have been implemented in response to a high number of conflicts (Gniadek and Kendall, 1998; Greenleaf et al., 2009). Although the exact effect of bear-safe containers is undetermined, they are widely used as a tool to reduce human-bear conflict (Spencer et al., 2007). Since bear management strategies have been met with varying success there is a need to determine what method and/or combinations of methods are the most effective in local communities (Baruch-Mordo et al., 2013; Hristienko and McDonald Jr, 2007).

Our goal is to use an individual-based model to quantitatively assess a wide range of bear management options with regard to their effectiveness at reducing the number of conflict bears, using the Resort Municipality of Whistler as a case study. The model explicitly simulates bear movement, foraging and human-bear interactions. The model also includes a simple representation of bear memory and its effect on foraging behaviour, following the work of Bracis and Mueller (2017). The model bears' interactions with the environment were compared under different bear management strategies in order to determine the most effective strategies for reducing human-bear interactions. We found that all bear management strategies reduced the probability of the model bear becoming a conflict bear (henceforth, probability of becoming a conflict bear) and the reduction achieved varies with the type of bear management strategy implemented.

## 2. Study area

Our study area was Whistler, British Columbia, Canada, which is located in the Coast Mountains approximately 120 km north of the city of Vancouver (50°07' N, 122°57' W). Whistler had a permanent human population of 11,854 in 2016 (Statistics Canada, 2017). This area

experiences extremely high tourist levels year round (summer and winter) (Kelly and Williams, 2007). In the summers (July–October) of 2000 and 2002, Whistler had 183,700 and 225,000 visitors, respectively (Needham et al., 2004). The area is home to many hospitality-focused venues, including recreation and accommodation (Kelly et al., 2006).

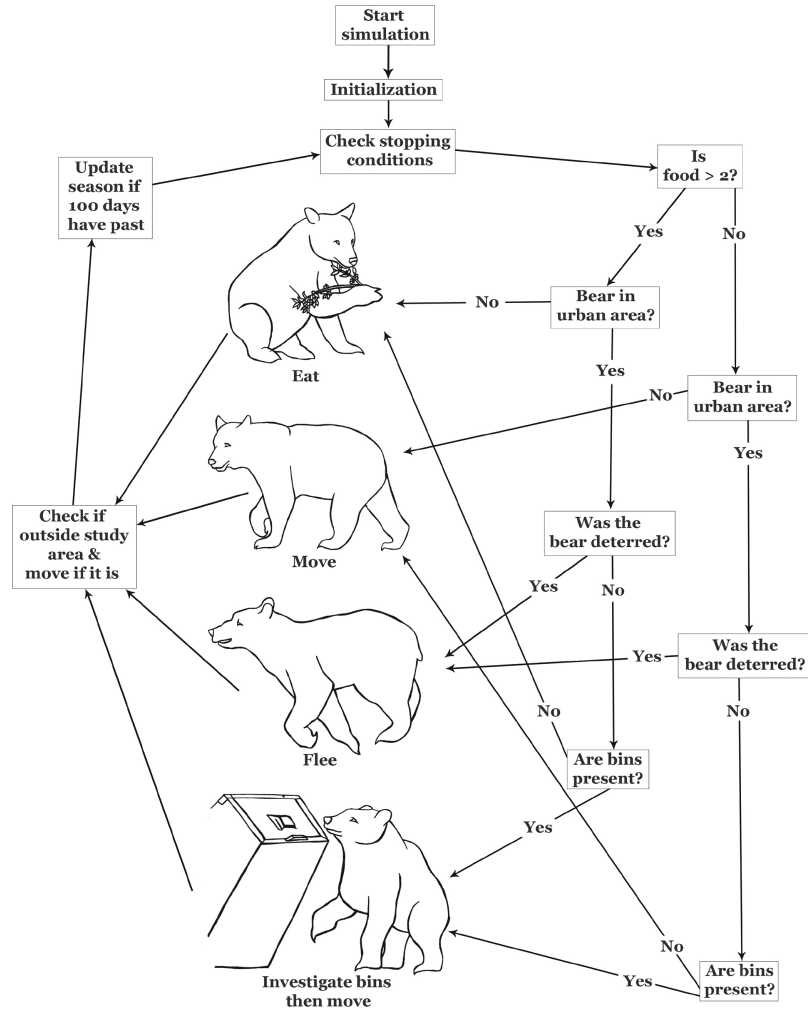
Approximately 100 black bears reside in the Resort Municipality of Whistler (Homstol, 2011). The area likely has high black bear population numbers due to high quality foraging in the area, which is enhanced by logging, ski hills, and other land management practices that result in the growth of herbaceous plants in the absence of trees (Appleton, 2006). Whistler has been previously found to experience both residential human-black bear interactions and visitor human-black bear interactions (Gore et al., 2006). Starting in 1998, there was a large effort to educate people on bear-safe practices (Baruch-Mordo et al., 2009) and the municipality is still working at reducing human bear interactions in the area (Resort Municipality of Whistler, 2016). With ample surrounding bear habitat and many bear attractants in the urban area, Whistler shares many characteristics with other small mountain communities in Western North America in terms of the challenges it faces in managing human-wildlife conflict.

## 3. Model structure and flow

We created an individual-based model of bear movement and human-black bear interactions in Whistler for the purpose of investigating the effectiveness of bear management strategies in reducing human-bear interactions (HBIs). The goal was to investigate several bear management strategies as well as to identify the best performing strategy and spatial configuration over which that strategy was implemented (see Section 3.3). This model applies the theoretical work of Marley et al. (2017) to a real, physical landscape. A full description of the model and its components are provided in Appendices A through G in the ODD format (Grimm et al., 2006, 2010). Here we provide an overview of the key model concepts.

The model landscape is divided into cells that are 25 m by 25 m. Each cell can have the following attributes: amount of natural bear food, amount of urban bear food, whether the bear is inside or outside the study area, urban or non-urban classification, and the bear's memory value for that location. The landscape includes both urban and natural environments, with humans integrated into the landscape. That is, humans are not modelled individually but rather as part of a cell's status (containing or not containing humans). Humans are not represented as individuals in the model since conflict bears don't directly interact with humans in most cases. Most conflict bears interact with the human landscape, and specifically with sources of human food on the landscape, such as dumpsters, backyard fruit trees, and other accessible human food waste. For this reason, we chose to incorporate humans in the model via modifications to the landscape that influence the availability of human food sources in the urban environment. We did explore the effect of actions that might be taken to deter a bear from entering the urban environment or from accessing human food. While, in the real world, such actions might be taken by individual humans, the simplest way to implement deterrence was via a deterrence probability that is location-specific. We did not see any reason to have individual human agents executing such deterrence in the model.

There is a single bear agent that interacts with the landscape. The bear moves through the landscape seeking food. We only consider the plant-based proportion of the bear's diet in the landscape, as vegetation makes up the majority of a bear's diet in this region (Pelton, 2003; Graber and White, 1983; Greenleaf et al., 2009). The model runs for 4800 hours total, or 200 days. The model run time thus encompasses the spring and summer seasons during which the bears are active and searching for food. The simulation is divided into two seasons, pre-berry and berry, and changes seasons at day 100. An overview of the model flow is given in Fig. 1.



**Fig. 1.** The processes the model follows this general process during run time. The bear has four major responses to landscape stimuli. For details on the effect on the memory and food matrices see Sections 3.1 and 3.2, and for details on changes in bear classification/stopping conditions see Section 3.4. All other processes are described in Section 3.

Bear management is defined as a purposeful action implemented in an urban cell with the intention of reducing human-black bear interactions. We focus on the bear management strategies (BMS) applied as well as the bear management configurations (BMC) which are the spatial distributions of the BMSs.

### 3.1. Bear memory

Animals that move large distances are reliant on their ability to remember information about their surroundings and experiences (Fagan et al., 2013). Since bears have high caloric demands and an ability to quickly retain information, they are able to consume and remember the location and quality of multiple food sources across a variety of habitats (Peine, 2001; Schoem, 1990). We assume that the model bears have an ability to learn and retain information about their landscape. Model bears, therefore, have a spatial memory that records attributes of the landscape.

This memory is represented as a spatial matrix having the same resolution and extent as the simulated landscape. The initial values in the memory matrix are the starting food values for the entire landscape. As a bear moves through a cell, it may increase or decrease its memory value for that cell depending on its experience at that location. For example, if good food (greater than 3) was found in an urban cell, the value is increased. If the bear is deterred from the urban cell, the value is decreased. Memory is later used to drive bear movement (see below),

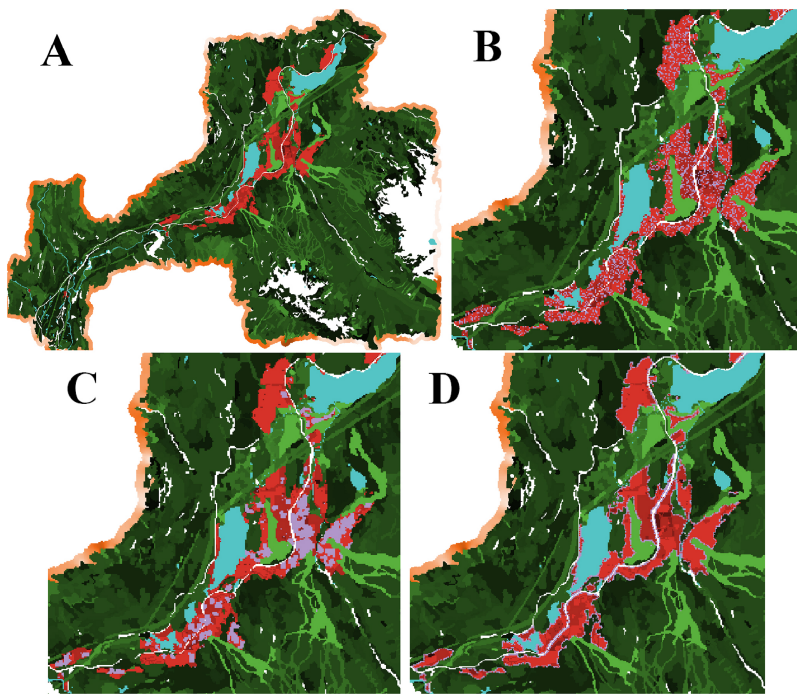
meaning that bears are more likely to move to cells that have high memory values. The purpose of the memory matrix is to incorporate the learning of the bear to the food availability of the urban landscape, while retaining previous knowledge of the natural landscape.

### 3.2. Movement

At each time step, the model bear selects a new cell to move to. Model bear movement is influenced by the food values and bear memory values of the landscape. Higher combined food and memory values correspond to higher probability of a bear selecting that cell. Before the bear chooses a new location, it considers several possible locations. These possible locations are either a short distance away if the food cell values surrounding the bear are high, or a long distance away if the food cell values surrounding the bear are low (see Section 5.2). The possible locations are determined using distributions fitted to the collar bear data step length (see Appendix A) and are converted to possible steps.

Once several potential steps are determined, the selection probability across all potential steps is calculated, with the result depending on the combined food and memory values in each potential destination cell. Mathematically, the selection probability is given by

$$P(S_{i,t}|F_{i,t}, M_{i,t}) = \frac{F_{i,t} + M_{i,t}}{\sum_{j=1}^n F_{j,t} + M_{j,t}} \quad (1)$$



**Fig. 2.** The study area bear food values for the natural landscape (green), the urban landscape (red), and the buffer zone (orange). Lighter shades correspond to high food values. Water areas are represented by blue. White represents no food values which can occur due to rocky surfaces, roads or no data available. Panel A shows the entire landscape; panels B–D show an enlarged view of the urban area. If a bear management strategy is implemented in an urban cell, that cell is coloured purple. (A) No management strategy, full landscape. (B) Random bear management configuration (BMC), zoomed in on urban area. (C) Cluster BMC, zoomed in on urban area. (D) Perimeter BMC, zoomed in on urban area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where  $S_{i,t}$  is the selection of cell  $i$  at time  $t$ ,  $F_{i,t}$  is the food value in cell  $i$  at time  $t$ ,  $M_{i,t}$  is the memory value in cell  $i$  at time  $t$ , and  $n$  is the total number of possible destination locations. The model bear will move only if the current cell food value is low (less than 3). Every time step spent in a given location reduces the food value of that cell by 1 unit, consistent with the idea that the bear is consuming the food resources in that cell.

If the bear enters an urban cell, there is a default probability (3.15%) that the bear is deterred. This probability was derived from human-bear interaction data within Whistler. If there is a deterrence event, the bear moves to a predetermined cell in a random direction to a cell not in the urban area and at a distance of  $(25\sqrt{2})r$ , where  $r$  is the number of cells the bear will run.  $25\sqrt{2}$  is the maximum distance within a single cell (since each cell is 25 m by 25 m) and by multiplying by the maximum distance we avoid the bear staying in the same cell in the time step when the bear is deterred. For each successive deterrence event,  $r$  is decreased by 1 and has an initial value of 5. In the case that there are no non-urban (natural) cells nearby, it will then choose any random cell.

Should a potential movement destination be selected outside of the study area, the bear will spend a randomly selected number of time steps outside of the study area. The bear will then be relocated to a randomly selected location that lies within the study area and is adjacent to the border.

### 3.3. Bear management

Bear management strategies (BMSs) are implemented on a selected number of the urban cells and in a specified spatial configuration. The model has three BMSs: (1) decreased availability of anthropogenic food, (2) increase in the probability of bear deterrence, and (3) installation of bear-safe bins.

Each BMS has a different effect on both the bear and the affected urban cell. Decreased availability of anthropogenic food (BMS #1) reduces the food value in the affected cell to the minimum value of 1. BMS #2 results in a 15% increase in the probability that a deterrence event occurs if the bear enters the affected urban cell. Installation of bear-safe bins (BMS #3) does not affect the food value of the cell but

instead denies the bear access to the food. This lack of access is modelled as a decrease in the bear's memory for that cell.

Any BMS that is implemented in the model is applied spatially under one of three Bear Management Configurations (BMCs). The model has three BMCs: (1) random, (2) cluster, and (3) perimeter. The random BMC will randomly assign the BMS to a selected number of urban cells (Fig. 2, image B). Cluster BMC creates groups of managed zones (Fig. 2, image C) within which the BMS is applied to urban cells. Finally the perimeter BMC applies the BMS to the cells at the urban/forest interface, i.e. the cells that border the natural environment (Fig. 2, image D).

### 3.4. Bear classification

Each simulation represents the foraging activity of a single bear and the simulation ends either at 200 days or when the bear is removed from the landscape, whichever occurs first. Bear removal from the landscape occurs if the bear has become a conflict bear (see below). Every time a bear enters an urban cell, one of three events occurs: the bear is deterred; the bear stays but doesn't find food; or the bear stays and does find food. Respectively, we call these events deterrence, habituation, and food conditioning. We record the total number of human-bear interactions (HBIs) of each type and use these numbers to determine the bear's classification as either a survival bear (remains on the landscape) or a conflict bear (removed from landscape) as shown in Fig. 3. Food conditioning is treated separately from habituation, as habituation (becoming accustomed to humans) can occur without a food reward and food conditioning may still occur in a bear that is nervous of people (Herrero, 1985; Herrero et al., 2005; Hopkins III et al., 2010). We consider food conditioning and habituation to be scaled, rather than binary, variables as shown in Fig. 3 (Herrero et al., 2005). Bears that are both habituated and food conditioned are more likely to become conflict bears (Mattson et al., 1992). Deterrence causes bears to become more wary of humans, and thus decreases habituation. In contrast, food conditioning in the model is irreversible, and can only increase. The simulation ends when either the bear lives through the entire season or is removed due to negative interactions with the human component of the urban landscape, i.e., becomes a conflict bear.



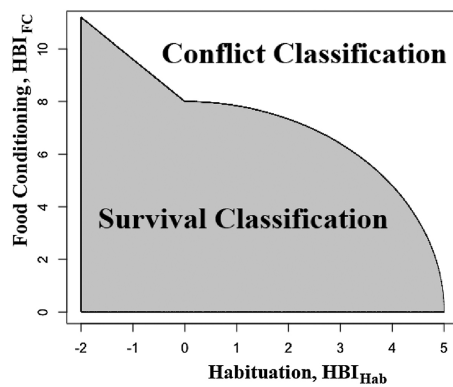


Fig. 3. Bear classification is determined as shown in this graph. Should the habituation or food conditioning level result in the bear being classified as a conflict bear the simulation ends. The shape of this graph was determined by classified provincial government documentation (Marley et al., 2017).

#### 4. Data

Four major data sets were used in the parameterization of this model; terrestrial ecosystem mapping (TEM) data (Cascade Environmental Resource Group Ltd., 2013), municipal zoning information (Resort Municipality of Whistler, 2015), black bear telemetry data (Homstol, 2011), and human-bear interaction records (Homstol, 2011). The telemetry data provided hourly GPS locations for 7 black bear individuals (one female, 6 males) over the years 2010 to 2011, with a 1 hour fixed time interval. All of the collared bears had a known history of being involved in human-bear conflicts, and so were in the process of becoming conflict bears and conditioned to human food sources (i.e., they were substantially habituated to humans and human food sources). The terrestrial ecosystem mapping (TEM) (Cascade Environmental Resource Group Ltd., 2013) provided a detailed survey of vegetation type across the study area, including information on the plant community, soil moisture, soil nutrient regime, and climatic characteristic covariates. These particular data were used because of the fine scale spatiotemporal aspects, as well as strong consistency in the goals of the proposed model, that is the bear data was collected to specifically include aversive conditioning/fleeing events and bear interaction with an urban landscape.

#### 5. Model parameterization

##### 5.1. Landscape

The data described in Section 4 were used to construct bear food availability maps for the natural food landscape in conjunction with expert consultation (our experts (SES and TH) are BC bear biologists and co-authors on this publication). The natural landscape food availability varies over the foraging period, and so for modelling purposes was split into two seasons, pre-berry and berry. These two seasons represent the expected natural bear food values before and during hyperphagia respectively. Seasonal natural food values were estimated by plant species and their ability to bear fruit, and were ranked for their relative estimated levels of natural forage supply on a scale of 1 to 5, where 5 is high. The value of any given plant resource varied with season. For example, areas of high herbaceous species concentration are of higher value in the pre-berry season, whereas berry bushes and fruit are of higher value in the berry season. Fig. 2 shows the relative natural food values for the pre-berry season in the green gradient (colour online).

The urban food environment was represented using municipal zoning information from the Resort Municipality of Whistler (2015). Urban bear food values for each zone were determined using a logistic

regression, reminiscent of a resource selection function (Johnson et al., 2006), as well as detailed food availability information for the individual zones (Resort Municipality of Whistler, 2015). The logistic regression was applied to the presence or absence of a real collared bear in each unique urban cell, that is a binary operator was assigned for each cell and was a 0 if a bear never visited the cell, or conversely a 1 if a bear had ever visited the cell. The model used for the regression is as follows,

$$P_{\text{bear}} = \exp \left( \sum_{i=1}^n \beta_i X_i \right) \quad (2)$$

where  $P_{\text{bear}}$  is the probability of the presence of a bear,  $n$  is the number of main municipal zones, and  $X$  is the municipal zoning type. Each resulting  $\beta$  coefficient was used as a relative base for each major zoning area (that is the values were normalized then multiplied by 5, thus placing the food value of each area on a scale of 1–5). The values for each bear were considered separately to identify any clear individual bias. The final urban food value was then assigned (between 1 to 5) based on the average value of the coefficient from all combined bears for that zoning type, as well as the known bear attractors in the area. For example, areas with restaurants or housing were given high values and areas with annex buildings or little vegetation were given low values. Fig. 2 shows the relative food values of the urban area with a red gradient (colour online).

##### 5.2. Movement

Model bear movement was calibrated using telemetry bear collar data (Homstol, 2011). The turn angle and step length distributions of bear movement trajectories were fitted to the GPS points using a least squares method. These distributions were then normalized to provide probability density functions (PDFs) for the selection of turn angles and step lengths for the model bears. The step length distribution for choosing an available cell depends on the food values surrounding the current cell; if the food values are high, the bear is more likely to stay nearby, while if local food values are low, the model bear will move farther away (Amstrup and Beecham, 1976). Therefore, we obtained two different probability density functions for the step length, depending on the food quality around the bear's current cell. That is, if the average surrounding food value is less than 3, the bear will use the long step PDF (minimum distance of 100 m), and if the average food values surrounding the bear are 3 or greater the bear will use the shorter step distribution (maximum distance of 100 m). Several possible steps are then selected using the appropriate probability density function and the final destination is selected as described above in Section 3.2 using the combined information from food availability and bear memory.

#### 6. Simulation experiments

Every possible combination of the three base BMSs (increased deterrence, decreased food, bear bins) was used in the simulations, giving a total of seven BMSs. With the 7 possible BMSs and 3 possible BMCs we had 21 possible Bear Management Programs (BMPs). Each BMP was applied to a given percentage of the urban area; five different values were tested (Table 1, third column). We thus had  $(7 \times 3 \times 5) + 1 = 106$  possible combinations to evaluate including the control (0%) (see Table 1). Each combination was replicated 20 times, except the control of 0% which was replicated 200 times. In total, 2300 simulations were performed. The bear classification and, if applicable, the time of death (time at which “conflict” status was achieved) were recorded at the end of each simulation.

#### 7. Statistical analysis

We applied survival analysis to the simulated data set using the

**Table 1**

List of tested values for each explanatory variable in the model. Every possible combination of these variables was simulated with 2300 replicates.

| BMC       | BMS                                                 | % Managed |
|-----------|-----------------------------------------------------|-----------|
| Random    | Decreased Food                                      | 0%        |
| Cluster   | Increased Deterrence                                | 4%        |
| Perimeter | Bear Bins                                           | 8%        |
|           | Decreased Food and Increased Deterrence             | 12%       |
|           | Decreased Food and Bear Bins                        | 16%       |
|           | Increased Deterrence and Bear Bins                  | 20%       |
|           | Decreased Food, Increased Deterrence, and Bear Bins |           |

“survival” package in R to obtain the probability of survival (hazard function), as well as the Kaplan–Meier curves for all covariates. We then used Cox’s proportional hazard to obtain the  $\beta$  coefficients, significance, and confidence intervals for the survival probability as influenced by the covariates. The response variables were the binary conflict status for the bear (0 for survival classification and 1 for conflict classification) and the time of conflict bear classification (i.e., the bear’s simulation runtime). The simulation end time is censored at 4800 hours. The covariates are management percent, BMC, BMS, and combined BMC and BMS.

## 8. Model validation

We validated the individual-based model of bear movement and resulting habitat selection by comparing the food values used by each of seven collared bears to the used food values for the simulated bears in both the pre-berry and berry seasons. The collared bear data used for validation were from the same seven bears used for model calibration due to limited data availability. The simulated bears were initialized in the same locations as the collared bears (i.e., the first recorded GPS point for each bear) and were allowed the same number of time steps as recorded for the respective collared bear. During validation, bears that were classified as a conflict in the simulation were not removed and instead allowed to finish their run. The food values for each location choice during simulation were recorded during run time. The collared bear locations were determined for both the pre-berry and berry food values separately on the landscape.

The frequency of use for each food value for the real and simulated bears is shown in Fig. 4. The correspondence is much closer for the berry season than for the pre-berry season. This discrepancy is likely due to the small sample size and the fact that most of the collared bear data was gathered after July during the berry season. We fit a Poisson

distribution using a generalized linear model to the bear food values with only a binary factor, denoting whether the bear was simulated or real (collared) as the independent variable. We found that there was a significant difference in the selected food values for the pre-berry season ( $p$ -value of  $9.41 \times 10^{-12}$ ), and no significant difference between the simulated bears and real bears for the berry season ( $p$ -value of 0.332).

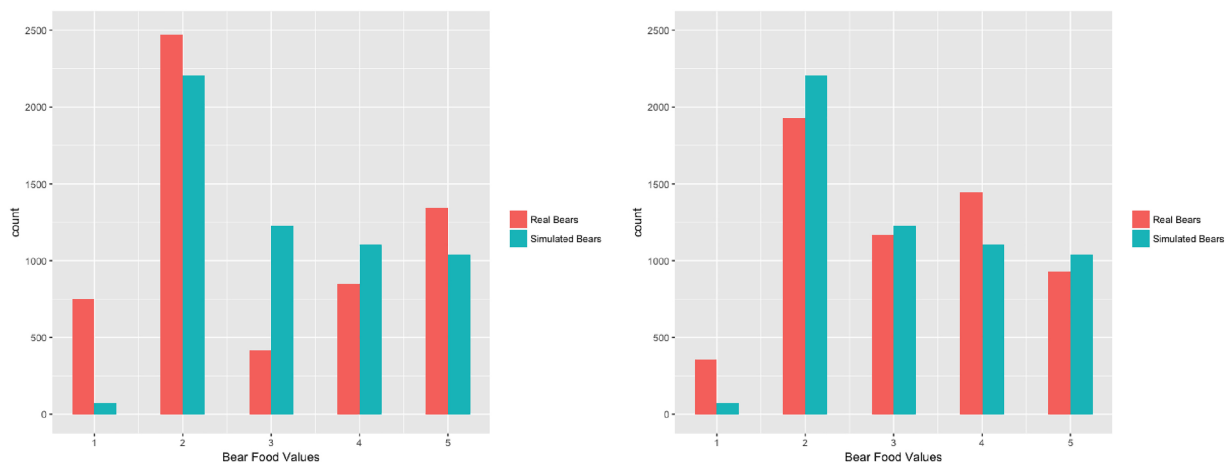
## 9. Results

For each combination of management percentage, BMC, and BMS, we recorded how long the bear survived in the simulation before becoming a conflict bear (here called survival time). Simulated bears tend to survive longer with higher management percents. Decreasing food availability (food BMS) also yielded the highest survival rates. Differences between the BMCs are not clear. Approximately 78% of all simulated bears were classified as conflict bears during the simulation time. We tested four possible models in the survival analysis. These are

1.  $h(t) \sim \text{BMC}$
2.  $h(t) \sim \text{BMS}$
3.  $h(t) \sim \text{Management \% (BM \%)}$
4.  $h(t) \sim \text{BMC} \times \text{BMS}$

where  $h(t)$  is the hazard function, Management % is the percent of the urban area on which a management strategy is implemented, and  $\text{BMC} \times \text{BMS}$  gives the 21 unique combinations of BMSs and BMCs. The  $\text{BMC} \times \text{BMS}$  term allows for the determination of the most effective combination of BMS and BMC. Table 2 shows the information from the summary tables of Cox’s proportional hazard. AIC determined that the best model for survival is Management %, followed by BMS and BMC respectively. The  $\exp(\beta)$  value of Management % indicates that if 100% of the urban cells are managed, survival will increase by 91%.

The BMC model in Table 2 shows that all BMCs had a significant impact on the survival of the bears, however the changes in survival are fairly similar between the three BMCs. In order of effectiveness they were random, cluster, then perimeter. The BMSs, however, are more diverse in their effect on survival, though all were found to have a significant impact. The most effective BMSs were decreased food availability and the combination of decreased food and increased deterrence. The least effective BMS was all three BMSs combined, followed by bins. The relationship between the BMSs appears to be non-linear, and applying multiple strategies in a single area can reduce the relative effectiveness of the other techniques, thus the BMS’s are not independent when used concurrently in the same location. For example,



**Fig. 4.** The use of landscape food values for the collared bears (red) and simulated bears (blue). The left graph corresponds to the pre-berry season, whereas the right graph corresponds to the berry season. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 2**

Summary of fitted hazard function using Cox's proportional hazard for three models on the survival of simulation bears. The confidence intervals were calculated with 95% confidence. Here BM% is the management percent, det is the increased deterrence BMS, food is the decreased food BMS, and bin is the bear safe container BMS. Significant results are indicated by a \* adjacent to the  $p$ -value for  $\alpha = 0.05$ .

| Model                   | Covariate | $\beta$ | $\exp(\beta)$ | Lower C.I. | Upper C.I. | $p$ -value              | AIC       |
|-------------------------|-----------|---------|---------------|------------|------------|-------------------------|-----------|
| $h(t) \sim \text{BMC}$  | Random    | -0.43   | 0.65          | 0.55       | 0.77       | $9.38 \times 10^{-7*}$  | 25801.99  |
|                         | Perimeter | -0.36   | 0.70          | 0.59       | 0.83       | $4.24 \times 10^{-5*}$  |           |
|                         | Cluster   | -0.38   | 0.68          | 0.58       | 0.81       | $1.25 \times 10^{-5*}$  |           |
| $h(t) \sim \text{BMS}$  | Det       | -0.41   | 0.66          | 0.54       | 0.80       | $3.82 \times 10^{-5*}$  | 25798.84  |
|                         | Food      | -0.47   | 0.62          | 0.51       | 0.76       | $2.42 \times 10^{-6*}$  |           |
|                         | Bin       | -0.26   | 0.77          | 0.63       | 0.94       | 0.00918*                |           |
|                         | Food, Det | -0.47   | 0.62          | 0.51       | 0.76       | $2.62 \times 10^{-6*}$  |           |
|                         | Food, Bin | -0.42   | 0.66          | 0.54       | 0.80       | $3.67 \times 10^{-5*}$  |           |
|                         | Det, Bin  | -0.43   | 0.65          | 0.54       | 0.80       | $0.228 \times 10^{-5*}$ |           |
|                         | All       | -0.25   | 0.78          | 0.64       | 0.95       | 0.01290*                |           |
| $h(t) \sim \text{BM\%}$ | BM%       | -2.38   | 0.09          | 0.04       | 0.19       | $2.04 \times 10^{-10*}$ | 25,780.53 |

the effect of bins is likely to be less prominent in areas with no food attractants, due to the bears being less likely to enter those areas even in the absence of bins.

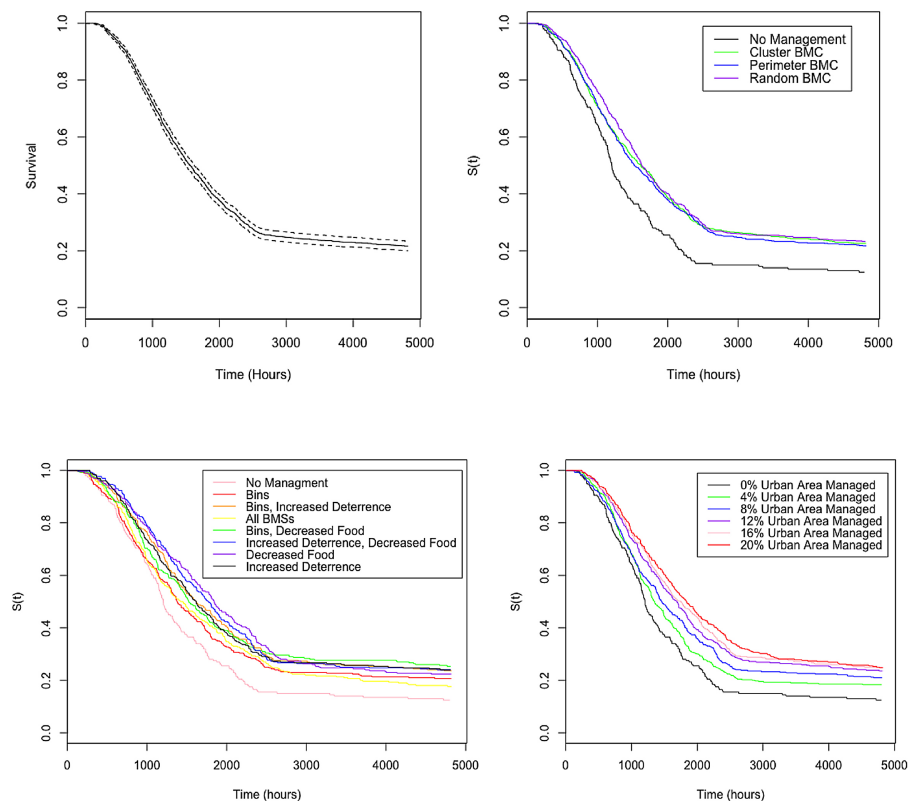
The Kaplan–Meier curves (Fig. 5) show that at time 1400, which corresponds to the change in season, there is an inflection in the slope of the line. The smaller slope indicates that bears have higher survival when the food quality in the natural environment is high. The curves for BMC, BMS, and Management % clearly show increased survival for all simulations that included management. The highest discernible difference is for Management %. It appears that, by the Kaplan–Meier metric, the most effective BMS is either decreased food or bins with decreased food, and the least effective BMS is either bins alone or all three BMSs combined. All three BMCs seem to have a comparable effect on survival.

It is clear that while all BMPs were useful in reducing the number of conflict bears, some programs were more useful than others. Fig. 6

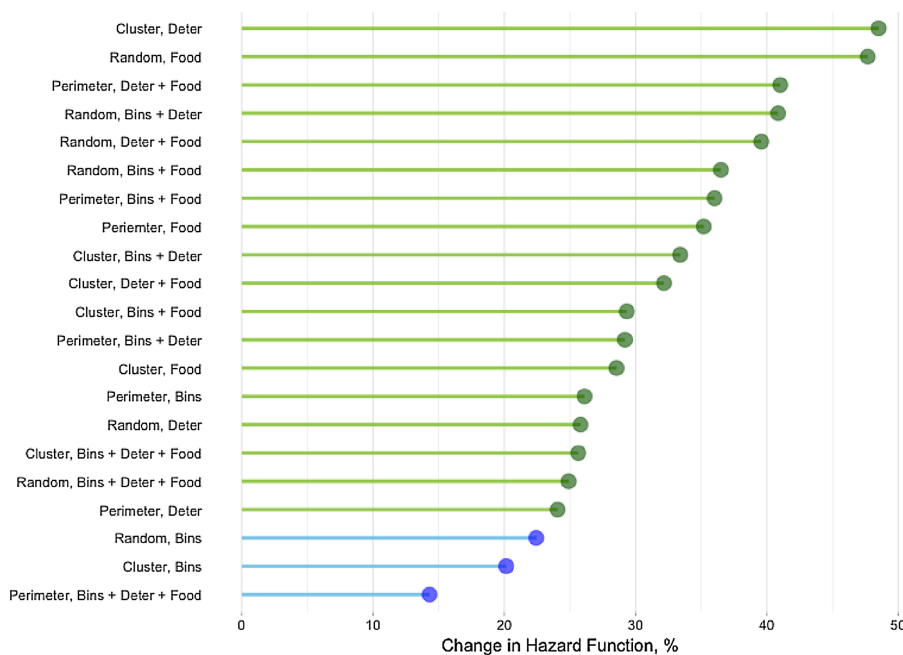
shows the unique combinations of BMSs and BMCs in order effectiveness from of most effective (top) to least (bottom). The most effective methods are cluster increased deterrence and random decreased food, with both of these increasing survival by approximately 48% and also decreasing food conditioning and habituation. The three least effective methods were random bins, cluster bins, and perimeter with a combination of all methods. The lower effectiveness of these last three methods may be due to the fact that bins still allow for attraction of bears to an area (and thus habituation) leading to the unexpected interaction of BMSs when they are used together.

## 10. Discussion

In general, preventative bear management is stronger and cheaper than reactive bear management (Riley et al., 1994; Hristienko and McDonald Jr, 2007) and our results indicate that the BMPs we



**Fig. 5.** Kaplan–Meier curves for the survival analysis on all 2300 bears. From top left clockwise, all bears combined survival, bear survival for each BMC survival, management % survival, and BMS survival.



**Fig. 6.** The changes in probability of survival were calculated from the results of Cox's proportional hazard,  $(\exp(\beta) - 1) * 100$ . Significant results ( $p$ -value < 0.05) are shown in green. In each BMP name, the BMC is listed first, followed by the BMS. Deter BMS refers to the BMS of increased deterrence, food refers to decreased food availability, and bins refers to installing bear-proof containers. The x-axis shows the change of probability of survival as compared to no bear management implemented. All values are positive and thus show an increase in survival. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

investigated may be viable options for the Resort Municipality of Whistler. While our results cannot be directly applied to other municipalities, we have demonstrated that using data from a specific community, it is possible to use our modelling approach to evaluate a wide range of BMPs inexpensively. Other communities can use this approach if they collect the necessary data (spatial data and collared bear data). If community-specific bear collar data is not available, our model, with Whistler bear step selection and resource selection functions, could be used in a community-specific landscape map to obtain preliminary results regarding BMP effectiveness in the new landscape.

Bears may alter their use of an urban landscape based on development density (Merkle et al., 2011) which may cause the spatial structure of management and management type to be adjusted depending on the qualities of each urban area (Johnson et al., 2015). Also, managing conflicts depends on the conflict types occurring and targeting areas of high known conflict may be important (Baruch-Mordo et al., 2008). Conflicts occur in a variety of landscape types (e.g. urban, recreational, agricultural, and back country) (Gore et al., 2006), which means that multiple management strategies should be considered depending on the nature of the conflict in each area (Decker et al., 2001).

Earlier work suggests that access to anthropogenic food should be prevented to avoid bears associating high calorie food rewards with humans (Herrero et al., 2005). Our modelling results support the empirical evidence that removal of attractants is the most effective strategy. However, this task is not always simple or possible (e.g. in National Parks, or in locations where there is a high density of tourists without a stake in preserving the local bear community) (Hopkins III et al., 2014). It is also thought that naturally occurring collapses in food productivity in the wilderness may drive bears to increase their use of urban areas (Baruch-Mordo et al., 2009, 2014; Johnson et al., 2015; Mattson et al., 1992) and thus lead to an increase in human-bear conflicts (Baruch-Mordo et al., 2008; Ryan et al., 2007; Zack et al., 2003). Our results indicate that survival is lower in the pre-berry season when natural food values are lower on average for the RMoW. Bears with low fat reserves are more persistent in their quest for food (Baruch-Mordo et al., 2013), which occurs as they enter hyperphagia behaviour. This phenomenon could be included in the model for future iterations, along with dynamic management strategies, to determine if the most effective BMS changes with time.

While this type of model could be quite useful for wildlife management planning applications, results should be considered with

caution due to inherent assumptions made. For instance, the model bears do not move continuously throughout the landscape but rather “jump” from one location to another at discrete time steps. These models should be used more as guidelines in conjunction with other considerations that are locally and biologically important. Also worth noting is our assumption that the people in an area will behave perfectly according to the BMS implemented. The assumption is clearly unrealistic in the absence of strong buy-in from the community and strict enforcement (Dunn et al., 2008). Our focus, however, is on the impact of the management on conflict bears, rather than the practical application. It is not known whether education is providing long-term solutions to preventing conflicts by altering human behaviour due to a lack of evaluation (Gore et al., 2006, 2008; Keane et al., 2008) and in some cases show little impact on human behaviour (Baruch-Mordo et al., 2011; Pienaar et al., 2015) and may not result in significant decreases in conflicts (Baruch-Mordo et al., 2011; Gore et al., 2006; Lackey and Ham, 2003). More proactive approaches to management may be necessary, for example through bylaws (Baruch-Mordo et al., 2011). While the presented model does not provide a solution to human cooperation, it can show the predicted effectiveness of successfully implemented management strategies.

In addition to quantifying BMS effectiveness, these types of models could be extended to include a cost-benefit analysis of various management strategies. Non-lethal actions often come with considerable financial and temporal cost (e.g. translocation, electric fencing, garbage removal, etc.), rendering many solutions infeasible (Rauer et al., 2006; Gore et al., 2006). Many methods are also viewed less favourably by public opinion (Hristienko and McDonald Jr, 2007). It is also necessary to balance human safety and conservation goals. For example, lethal management is not sustainable for grizzly bear populations, and could become a conservation issue for black bears as well (Honeyman, 2008). Knowing beforehand the expected decrease in conflict bears to total management cost could prove invaluable in sensitive situations. While non-lethal management does not provide a panacea for human-bear conflict, when used properly and optimally it can improve the chances of successful co-habitation between wildlife and humans in high quality habitat (Reddix and Forsyth, 2006; Homstol, 2011).

## 11. Conclusion

While population-level models of bears are considered to be



insightful (Miller, 1989; Lewis et al., 2014), individual-level models can also provide unique insight into other questions concerning bears, such as the ones addressed here on the management of human-bear interactions. For instance, it appears that it is optimal to not apply every possible management strategy in one location but rather focus on the most effective possible method in any given location. Models allow for relatively inexpensive exploration of a wide range of different hypothetical situations, as we have done here. These hypothetical situations do not have to be limited to bears, and could be applied to any species with human-wildlife conflict. Individual-based models with human-animal interactions could also be useful for other species, as it is expected that the increase of urbanization will lead to more human-wildlife conflict in general (DeStefano and DeGraaf, 2003; Shocat et al., 2006).

## Appendix A. Purpose

The purpose of this model was to explore the effectiveness of bear management strategies, and the spatial configuration of these strategies, in reducing the number of conflict bears/HBLs for a given community. The model extends the prototype of Marley et al. (2017) to a real landscape containing a small urban community for which bear movement data was available. In particular, we aim to quantitatively compare the effectiveness of several possible bear management strategies when implemented over a summer season in the Resort Municipality of Whistler (RMoW), in British Columbia, Canada.

## Appendix B. State variables and scales

For this model there are two types of variables; patch (landscape, human presence) and individual (bear). Each variable describes their states. Both types contain variables that have static and dynamic properties. Details are given in this section and in Appendix G.

The landscape is the set of patches, and is set up as a grid of square cells, with each cell being 25 m by 25 m. The RMoW is not a perfect square (Fig. 2), thus leaving many cells with no information. The total number of cells with information is 295,107, giving a total area of 184,441,875 m<sup>2</sup>. We used absorbing boundary conditions that will relocate the individual (see Appendix G.4). The bears are simulated individually, therefore group dynamics between bears are ignored. The temporal resolution is hourly time steps, running over 200 days. The seasons change on the hundredth day.

### B.1 Patch variables

The patch variables are derived from geospatial data for the landscape for the RMoW. These patches were made from Terrestrial Ecosystem Mapping (TEM) data gathered for ecosystem surveillance of Whistler (Cascade Environmental Resource Group Ltd., 2013). These patch variables include the natural landscape, the urban landscape, the buffer zone, the estimated bear food availability for the natural landscape, the estimated bear food availability for the urban landscape, and the bear memory. Each component is described respectively below.

#### B.1.1 Natural landscape

The natural landscape,  $\bar{L}$ , is the physical characteristics of the study area, which includes plant communities, soil moisture, soil nutrient regimes, and climatic characteristics (Cascade Environmental Resource Group Ltd., 2013). These ecosystem types are classified by the Biogeoclimatic (BEC) units to a sub-variant level. The TEM data was constructed from terrestrial ecosystem mapping (TEM) data. The data was collected with photo typing on alternate 1:15,000 colour aerial photos from 1994, giving implied spatial resolution of 75 m by 75 m (0.6 Ha) for a minimum identifiable homogeneous region.

By using ArcGis, the exact coordinates of the features was determined and mapped together through shape files. In the natural landscape there are four distinct zones, forest (or vegetative cover), high mountain (rock face), water (lake, river and stream), and urban recreational (ski hills, camp grounds, etc.). The forest has natural food sources for the bears and the landscape information was used in the assigning of food values (see Appendix B.1.3). The high mountain zone does not have food available for bear consumption. Cells surrounding water areas have high food values. The urban recreational areas were given food availability values in the same way as the forest zones. Altitude was not considered as an important factor in bear movement for this stage of the model, however this may be adjusted in the future.

The natural landscape variable is defined in the above method and classifies each cell as a bear habitat type. Each habitat type has a numerical value as determined by the report by Cascade Environmental Resource Group Ltd. (2013). There are three bear habitat type files that make up the natural landscape.

#### B.1.2 Buffer zone

The buffer zone,  $\bar{B}$ , represents the unknown landscape beyond the scope of the available TEM data. Patches bordering the study area have a buffer zone Boolean value of  $T$ . The buffer zone has food,  $\bar{B}^{\text{Food}}$ , and memory,  $\bar{B}^{\text{Memory}}$ , components associated with each location. The memory is set as equal to  $\bar{B}^{\text{Food}}$  and both remain static through the simulation. The food value is set by

$$\bar{B}_{i,j}^{\text{Food}} = \frac{\sum^K (\bar{B}_{a,b}^{\text{Food}} + \bar{N}_{a,b})}{n} \quad (\text{B.1})$$

where  $(a, b)$  are the  $(x, y)$  coordinates of the neighbouring cells with  $\bar{B}_{a,b}^{\text{Food}} > 0$  and  $\bar{N}_{a,b} > 0$ ,  $K$  is the set of all neighbours to cell  $(i, j)$ ,  $\bar{B}_{a,b}^{\text{Food}}$  is the buffer food value of cell in location  $(a, b)$ ,  $\bar{B}_{i,j}^{\text{Food}}$  is the food value of the buffer cell that is to be determined,  $\bar{N}_{a,b}$  is the natural food value of the cell in location  $(a, b)$ , and  $n$  is the number of cells in  $K$ .

## Acknowledgments

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### B.1.3 Natural food

The natural food layer,  $\bar{N}$ , is the bear food availability values for the non-urban area only. The food values are between 1 and 5, with 1 being the lowest quality and 5 being the highest. The natural food consists only of the vegetative part of the diet, we ignore prey values for simplicity. We can argue this since the majority of a bear's diet in this type of environment consisted of vegetative sources (Graber and White, 1983; Greenleaf et al., 2009).

### B.1.4 Bear memory

The bear memory variable layer,  $\bar{M}$ , contains the bear's learned attractiveness of an area. We assume that the bear remembers the attractiveness of its physical environment and their location as it traverses the landscape. Memory values are either 1, 2, 3, 4 or 5, with 1 emulating a memory of a low quality area and 5 emulating a memory of a high quality area. To start, the memory equals the initial food value, thus assuming that the bear knows the landscape from previous years and there were no changes. Values for the memory are affected by interactions with the urban environment (see Appendix G.5). The memory for any cell in the natural landscape never changes (either positively or negatively) during a season, in order to account for a bear's previous experience in that cells and its tendency to return to areas that it has previously visited. However when the seasons change the memory values in the natural landscape are reset.

### B.1.5 Urban landscape

The urban landscape includes all non-recreational urban classified areas from Bylaw 303 for the RMoW (Resort Municipality of Whistler, 2015). The urban area is considered on a cell scale only, that is there are no individual humans in the model. We ignore the effect of hikers, campers, skiers, etc. for simplicity. This way people are integrated into the environment itself. The urban landscape variable has a Boolean value of T when the cell is urban, and F when it is not.

### B.1.6 Urban food

The urban food variable,  $\bar{U}$ , is the bear food availability on urban cells. These food values do not change when the seasons change. To assign urban food values, we applied a logistic regression via a generalized linear model to the presence or absence (1 or 0) of the collared bears and used the general municipal zones for the RMoW as the only covariate. Each categorical zone was then associated with a coefficient value. These were then scaled between the values of 1 and 5 and served as a base level for the general zoning groups. We then went through all groupings and assigned them a food value based on the scaled coefficient values. The increases or decreases were based on the building allowances in each zone. For example, if a food source (ex. fruit trees, restaurant, housing, etc.) was present in the area then the food value would be increased.

### B.1.7 Urban vigilance

The urban vigilance variable is the probability that a bear is deterred when it enters the urban area. We are not assuming the method for which a bear is deterred, but rather the probability that a successful deterrence event occurs. Deterrence could be accomplished by a trained professional that has been notified of bear presence for example. The base value for each urban cell,  $v_{\text{initial}}$ , is 3.15% which was calculated from

$$\frac{\sum (\text{number of conflicts})}{\sum (\text{time in urban area})} = \frac{88}{2797.125} \quad (\text{B.2})$$

The vigilance increased with one of the management strategies (see Appendix G.6.1) to  $v_{\text{increased}}$ .

## B.2 Bear variables

Bear variables are the variables associated with each individual bear. These variables include conflict status, buffer status, deterred status, deterrence radius, habituation level, food conditioning level, turn angle of next step, step length of next step, and current spatial coordinates ( $i, j$ ). Each of these change throughout model runtime.

The conflict status is a binary value indicating whether the bear has become a conflict bear (see Appendix C.1). The buffer status and deterred status are also binary values, indicating whether the bear has entered the buffer zone (see Appendix B.1.2) or if the bear is currently being deterred (see Appendix G.5.3). For each deterrence the bear is deterred a shorter distance and the deterrence radius keeps track of the distance they will run (see Appendix G.5.3). Both the habituation level and the food conditioning level keep track of the impact that human-bear interactions have on the bear (see Appendix G.5). The turn angle and step length of the next step are determined based on the surrounding and current cell food and memory values (see Appendix G.3). The  $i$  and  $j$  location values give the coordinates for the current bear position.

## Appendix C. Process overview and scheduling

The bear movement and behaviour decisions are shown in Fig. C.7. This diagram also shows the overall model processes, including the start and stop. The model begins on a day during the foraging season set by the user and finishes once the bear is either classified as a conflict or survival bear (see Appendix C.1).

### C.1 Stopping conditions

The model finishes running when the bear agent is classified as a survival bear or a conflict bear. A survival bear lives throughout the entire simulation without being classified as a conflict bear (200 days). A bear becomes a conflict bear when it becomes too food conditioned and/or habituated as by the definition in Fig. C.8. This follows the same outline as in Marley et al.'s paper (Marley et al., 2017), however due to hourly time steps instead of the previous 12 hour time steps, each human-bear interaction will only increase food conditioning or habituation by 1/12.

Two evaluations are used for determining if the bear is a conflict bear, one for  $HBI_{Hab} < 0$  and the other for  $HBI_{Hab} \geq 0$ . In either case, if the bear's food conditioning level ( $HBI_{FC}$ ) is above  $FC_{max}$  then it is classified as a conflict bear and the simulation stopped. If the bear has a  $HBI_{Hab} < 0$ , we use

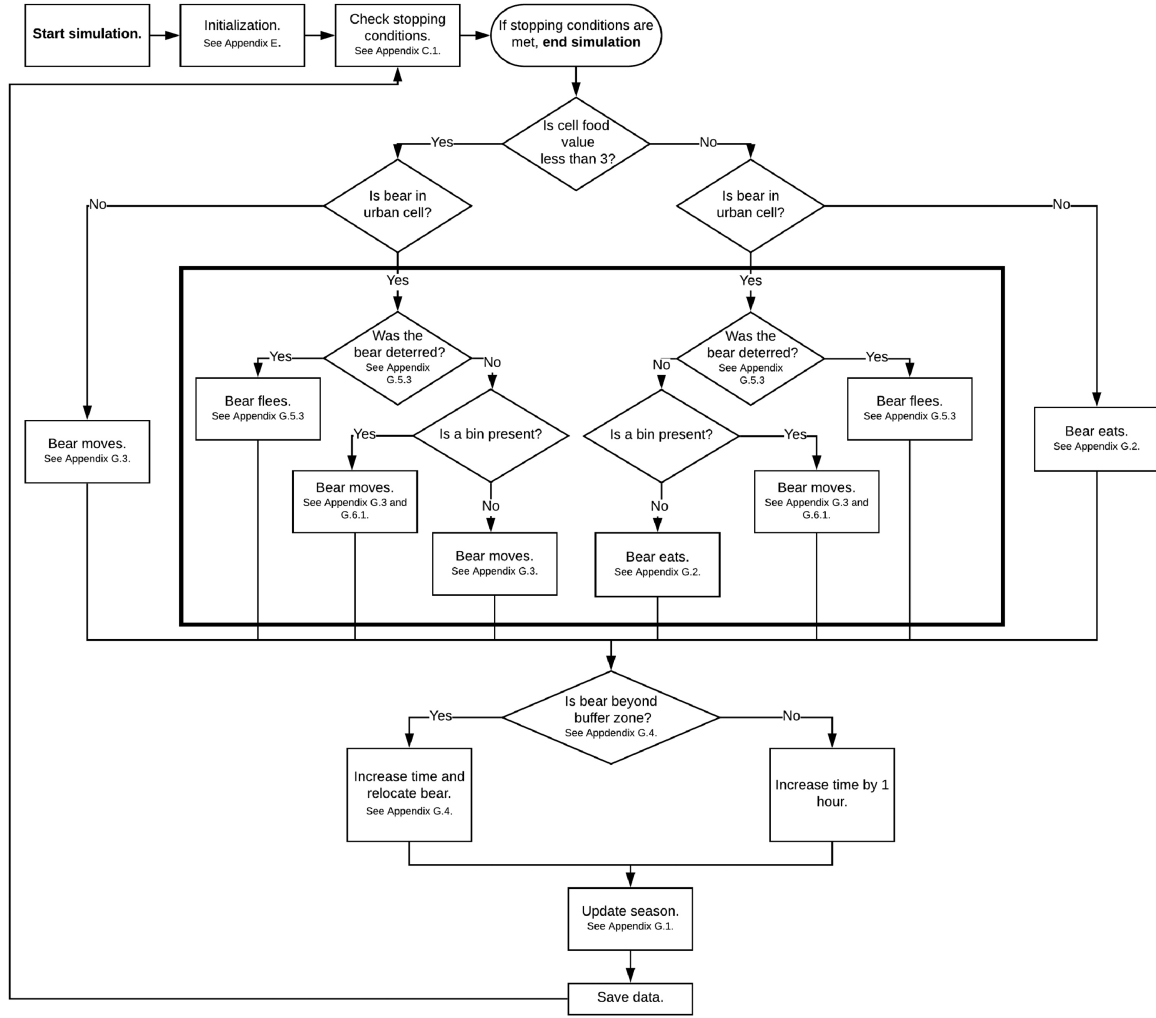


Fig. C.7. The processes the model follows during run time. All processes within the box outline occur in the urban environment.

$$FC_{\max} = -\left(\frac{8}{5}\right)HBI_{Hab} + 8 \quad (C.1)$$

to determine if the bear has become a conflict bear. However if it's the case that  $HBI_{Hab} \geq 0$  we use

$$FC_{\max} = \sqrt{8^2 \left(1 - \frac{HBI_{Hab}^2}{5^2}\right)} \quad (C.2)$$

These two equations describe the shape of the curve in Fig. C.8 for  $HBI_{Hab} < 0$  and  $HBI_{Hab} \geq 0$  respectively. The values for the food conditioning and habituation maximums were determined through expert consultation with conservation officers and do not necessarily represent real determinations exactly.

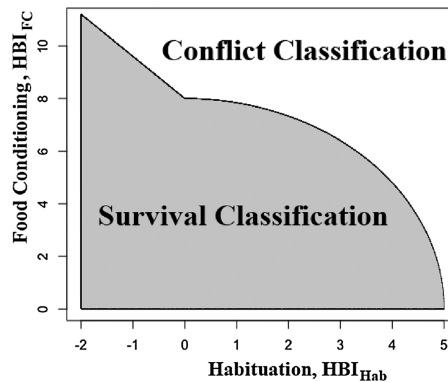


Fig. C.8. Bear classification is determined as shown in this graph. Should the habituation or food conditioning level result in the bear being classified as a conflict bear the simulation ends.

## C.2 Time

The model time,  $t$ , runs for an entire summer season from start to finish. We assume that it last 200 days, with a seasonal change occurring at day 100 (see Appendix G.1). The bears move on an hourly scale, thus there are 4800 time steps total.

## Appendix D. Design concepts

**Emergence** We only consider individual bears, thus there is no emergent group dynamics.

**Adaptation** The bears adapt to the landscape through their memory. They learn the location where they previously found high food values, as well as areas where they previously experienced a deterrence event or bear safe container. The memory influences the probability of visiting a location. For each deterrence event the bear runs a shorter distance away.

**Fitness** Fitness is not modelled explicitly.

**Prediction** The model predicts the change in bear survival when management strategies are implemented.

**Sensing** The bears can perceive the surrounding environment and use the perception in conjunction with memory to make movement decisions.

**Interaction** There are no interactions between individuals, however the bear interacts with the environment.

**Stochasticity** The bear agent's initial position is set randomly. There is randomness in the initialization for the random BMS (see Appendix G.6.2), the time adjustment and relocation of a bear when it enters the buffer zone, bear movement (turn angle, step length, and selection), and deterrence probabilities.

**Collectives** Only one individual is simulated at a time, thus no collectives form.

**Observation** The data is collected at the end of each individual simulation run and stored in a.csv files. These include bear status, time of conflict classification, covariate values, habituation level ( $HBI_{Hab}$ ), food conditioning level ( $HBI_{FC}$ ), number of deterrences ( $HBI_{Det}$ ), and the food consumption per time step. The overall results as well as the information per time step are saved.

## Appendix E. Initialization

The first step of the initialization process is file uploading and creation of the landscape matrices. The.csv files containing the landscape information (see Table B.3) is uploaded before the simulation starts. The natural and urban food patches are pre-initialized before simulation start to save time. The bear management strategy (BMS, see Appendix G.6.1) of the run is then applied for the chosen bear management configuration (BMC, see Appendix G.6.2). The bear is initialized in any random non-urban cell. All other initial parameter values (see Table B.4) are set to their initial values. The values are redefined as the bear interacts with the landscape and depending on the interactions, the new values are associated with the patch.

**Table B.3**

Variables associated with the environment. Each matrix is considered to be a layer in the model landscape.

| Patch variables            |                    |                    |
|----------------------------|--------------------|--------------------|
| Variable name              | Symbol             | Values             |
| Natural landscape          | $\bar{L}$          | Categorical string |
| Buffer zone                | $\bar{B}$          | 0 or 1             |
| Buffer zone food           | $\bar{B}^{Food}$   | 1–5                |
| Buffer zone memory         | $\bar{B}^{Memory}$ | 1–5                |
| Natural food               | $\bar{N}$          | 1–5                |
| Bear memory                | $\bar{M}$          | 1–5                |
| Urban landscape            | $\bar{Z}$          | Categorical string |
| Urban food                 | $\bar{U}$          | 1–5                |
| Urban vigilance, initial   | $v_{initial}$      | 3.15%              |
| Urban vigilance, increased | $v_{increased}$    | 18.5%              |
| Time                       | $t$                | 1–4800             |
| X Coordinate               | $i$                | 0–842              |
| Y Coordinate               | $j$                | 0–667              |

**Table B.4**

Variables associated with the individual bears. All these values may change during run time.

| Bear variables          |             |               |            |
|-------------------------|-------------|---------------|------------|
| Variable name           | Symbol      | Initial value | Range      |
| Conflict status         | $\delta$    | 0             | 0 or 1     |
| Buffer status           | $\gamma$    | 0             | 0 or 1     |
| Deterred status         | $HBI_{Det}$ | 0             | 0 or 1     |
| Deterrence radius       | $r$         | 5             | 0 to 15    |
| Habituation level       | $HBI_{Hab}$ | 0             | –2 to 5    |
| Food conditioning level | $HBI_{FC}$  | 0             | 0 to 11.2  |
| Turn angle              | $\theta$    | 0             | 0 to 360   |
| Step length             | $s$         | 0             | 0 to ~2000 |
| Current X coordinate    | $i$         | Random value  | 0 to 842   |
| Current Y coordinate    | $j$         | Random value  | 0 to 667   |



## Appendix F. Input

The landscape files that are inputted into the model during initialization are the landscape, the buffer zone, the natural food, and the urban food. All other model parameter are preset into the model (for example the default probability of deterrence,  $v_{\text{initial}}$ ).

## Appendix G. Submodels

### G.1 Seasons

We assumed two distinct phenological seasons, a pre-berry and berry based on the vegetation portion of the bear's diet. The season changes halfway through the simulation at day 100. The pre-berry season is considered the time from bear emergence in the early spring to mid-summer (approximately mid-July) where the herbaceous foliage, such as grass and green shoots, is the bear's primary diet (Graber and White, 1983; Greenleaf et al., 2009). The berry season is the time from mid-summer till the bear dens for the winter. During this season, calorie-rich berries, nuts and seeds are the bear's primary diet. Some plant species, such as tubers and manicured features, where the edible parts persist through both seasons are held constant across both season.

### G.2 Food consumption

A higher food values corresponds to a higher probability that a bear will go to that cell (see Appendix G.3). If the bear stays in a cell for a time step, the food values in that cell is reduced by 1 to imitate foraging. A bear will stay in a location until the food is less than 3. When the season changes to berry, the food values are reset to the berry season values, that is they are not affected by the bear's foraging in the previous 100 days.

### G.3 Movement

The bear movement in the model is based on collared black bear data in Whistler, BC from a study completed by Homstol (2011). Telemetry point data used for model calibration included 7 of 9 available individuals due to overlap with available spatial landscape data. Of the 7 individual bear's data used, 6 were male and 1 was female. The GPS collars were from Lotek (Nemarket, Ontario, Canada), 4400S model and reported location data in one hour intervals. The bears that were collared had a known history of being conflict bears or exhibited potential for conflict behaviours and because of this the telemetry data is biased towards the urban areas of the RMOW and tend to choose urban areas or adjacent to urban areas over the natural landscape.

Geospatial Modelling Environment was used to calculate the step length and turn angle of each collared bear's time series data points. From these calculations, statistical distributions (histograms) were generated (see Figs. G.9 and G.10). Turn angle is calculated as the clockwise rotation of angle between the line segment of telemetry point at the given time step and the line segment of telemetry point of the previous time steps. Turn angle distribution is a bimodal distribution with peaks at zero or 360 degrees and 180 degrees. Due to equal weighting of the distribution from 0 to 180 and from 180 to 360, it is assumed that the bears do not distinguish between left and right. They appear to have a tendency to travel either straight forward to turn around and travel back the way they came. The convenient shape lead to mirroring the distribution onto itself about the 180 degree mark, and making one distribution from 0 to 180 degrees (see Fig. G.9).

The step size was calculated as the straight-line distance between telemetry points at the given time step and the previous time step. When the step lengths were compiled into a histogram for all collared bears, the resulting distribution was an exponential distribution with a high frequency of short distance step lengths and a lower frequency of higher distance step lengths. These step lengths were normalized by time-step for an hourly movement length. The exponential distribution is assumed to correspond to frequent short distance foraging movements and less frequent longer distance relocation movements. We broke the distribution into two parts; short step size and long step size. A short step size is defined as being below 100 m and a long step size is any step size greater than 100 m (see Fig. G.10).

Several available patches are found using the distributions found above. Once they are found, the likelihood that a patch is chosen is determined by

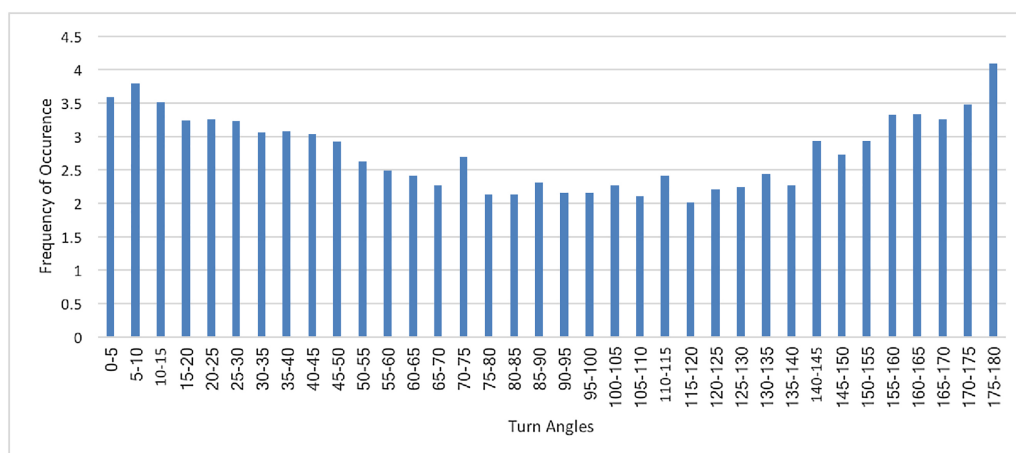
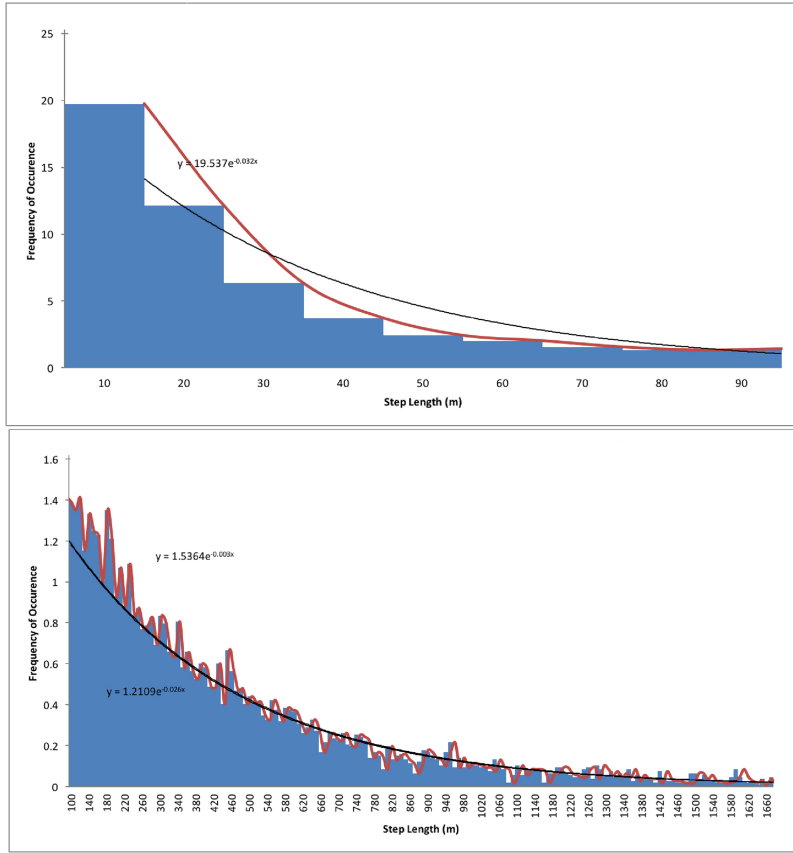


Fig. G.9. Histogram of collared bears turning angles. This distribution is used for the probability distribution function for the model turning angles.



**Fig. G.10.** Histograms of collared bears step lengths with lines of best fit. The top histogram shows the short step length distribution which is used when the model bear is surrounded by high food values, and the bottom histogram shows the long step length distribution which is used when the model bear is surrounded by low food values. These distributions are used for the probability distribution functions for the model step lengths.

$$P(S_{i,t}|F_{i,t}, M_{i,t}) = \frac{F_{i,t} + M_{i,t}}{\sum_n F_{j,t} + M_{j,t}} \quad (\text{G.1})$$

where  $S_{i,t}$  is the selection of cell  $i$  at time  $t$ ,  $F_{i,t}$  is the food value in cell  $i$  at time  $t$ ,  $M_{i,t}$  is the memory value in cell  $i$  at time  $t$ , and  $n$  is the set of all possible destination locations  $j$ . The location where the bear moves is then determined by a “roulette wheel” technique using a uniform random variable. The patch cumulative sum that is closest to the random variable is the new location.

#### G.4 Buffer zone conditions

Should the bear leave the study area it enters the buffer zone. In order to account for the chance that the bear may be gone for more than one time step, the following equation is used to create a random amount of time passing,

$$f(t) = 1 - \exp(-0.113t), \quad (\text{G.2})$$

where  $t$  is the time that passes and  $f(t)$  is the probability density function. The maximum time that can pass is 124 h. The bear is then placed in a random location adjacent to the buffer zone. The values for Eq. (G.2) were found from the times that the bear left the study area and were fit using the method of least squares.

#### G.5 Human-bear interactions

Human-bear interactions (HBIs) occur every time the bear enters the urban area. We classified three different types of human bear interactions; habituation, food conditioning, and deterrence. The details of these interactions are described in the following sections.

##### G.5.1 Habituation

Habituation,  $HBI_{Hab}$ , is here defined as the acclimatization to urban areas themselves but not anthropogenic food. In the model, habituation emulates a bear becoming less wary of humans without experiencing a food reward. In the model, habituation occurs when the bear enters an urban cell that has a urban food value less than 3 and the bear is not deterred. When a habituation event occurs, the  $HBI_{Hab}$  increases by 1/12. This type of HBI has no effect on the memory, either positively or negatively.

##### G.5.2 Food conditioning

Food conditioning,  $HBI_{FC}$ , is here defined as the exposure and consumption to high calorie anthropogenic food sources. In the model, food conditioning emulates a bear using human food sources and experiencing the reward associated with human high calorie waste and property (gardens, fruit trees, bee hives, etc.). In the model, food conditioning occurs when the bear enters an urban cell that has a urban food value equal to or greater than 3 and the bear is not deterred. The memory value for that cell is then increased by 1, and the  $HBI_{FC}$  increases by 1/12. The urban food value in that cell decreases by 1.

### G.5.3 Deterrence

Deterrence,  $HBI_{Det}$ , is here defined as any successful deterrence event that results in a bear fleeing an urban area. In the model, deterrence occurs when the bear enters an urban cell and if a random number is below the  $v$  value for that cell, either  $v_{initial}$  or  $v_{increased}$  if the decreased deterrence BMS is applied (see Appendix G.6.1). Once deterred, the bear runs to a random cell, to a cell in the natural environment at a distance of  $25(r)(\sqrt{2})$ , where  $r$  is the deterrence radius. However if there are no natural cells available at that distance the bear moves in a random direction at the same distance as described before. For each successive deterrence  $r$  is decreased by 1. Deterrence results in  $HBI_{Det}$  being changed to 1 for one time step, decreases the memory value in the cell that deterrence occurred by 1, and decreases the bear's  $HBI_{Hab}$  by 1/2.

## G.6 Management

### G.6.1 Bear management strategies

The model bear management strategies (BMSs) are the methods applied to urban areas in an attempt to decrease the number of conflict bears that occur. We do not focus on how these methods are implemented but rather the effect of a successful implementation of the strategies. The three methods we incorporate are decreasing urban food, increasing deterrence, and implementing bear safe containers.

Decreasing the urban food reduces the urban food value in that cell to the minimum value of 1. In practical application, this would be the removal of all attractants in an area. Increased deterrence is the increase of the  $v$  value by 15%. The 15% increase in deterrence was estimated by consulting experts. This increase results in the increase in the probability that a bear is deterred when it enters the urban area. Implementing bear safe containers does not affect the food value but instead denies the bear access to food. If the bear enters a cell that has a bear safe container, the cell's memory value decreases by 1,  $HBI_{Hab}$  increases by 1/12, and the bear leaves.

### G.6.2 Bear management configurations

The model bear management configurations (BMCs) are the potential spatial patterns of the implemented BMSs. We focus on three possible BMCs; random, perimeter, and cluster. The random BMC chooses random cells through the urban area to implement the BMSs, with no bias. The perimeter BMCs implements BMSs first in cells next to the natural environment, and then subsequently as close to the natural environment as possible. The cluster BMC randomly chooses cells to manage around a set of fixed random cells. This creates clusters of managed areas that change in size based on the percent that is selected to be managed. The number of cells chosen are the set percentage as determined per the simulation initialization.

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